

Name _____

Date _____

Period _____

Graphing Calculator Scavenger Hunt

(Reset Calculator- 2^{nd} $+$ 7 : $\downarrow$ Reset 1 : All RAM 2 : Reset Calculator Screen reads RAM Cleared)

1. Press 2^{nd} $+$ ENTER What is the ID# of your calculator? _____
2. For help, what website can you visit? _____
3. What happens to the screen when you push 2^{nd} $\uparrow$ over and over? 2^{nd} $\downarrow$ over and over?

4. Type the number 1234, then ENTER . You should see the same number in the display. Now type 9,876, 543,210,987,654, then ENTER
➤ Is the identical number displayed? _____ What is the largest number of digits that can be entered and still appear the same after you hit ENTER ? _____
5. $\wedge$ is called the "carat" button, and is used to raise a number to a power. Find $6^5 =$ _____. To square a number use x^2 What is 56^2 ? _____. To cube a number, press MATH and select option 3. What is 36^3 ? _____
6. What happens when you press 2^{nd} LOG 6 and ENTER keys are pressed? _____
7. Press 2^{nd} Y= to access the [STAT PLOTS] menu, how many stat plots are there? _____. Which option turns the [STAT PLOTS] off? _____
8. Press STAT which option will sort data in ascending order? What do you think will happen if option 3 is selected? _____
9. What letter of the alphabet is located above $\div$? _____. What would you press to get that letter? _____
10. To get the calculator to solve the following problem $2\{3 + 10/2 + 6^2 - (4 + 2)\}$, what do you do to get the { and }? _____ The answer to the problem is _____.
➤ Retrieve the expression you typed by pressing 2^{nd} ENTER . Using our arrows, change the { and } to (and). Does it make a difference which parenthesis were used? _____ Why might { and } be used? _____
11. To solve a problem involving the area and/or circumference of a circle, which calculator key(s) would you most likely use? _____
➤ Given the area of a square, which calculator key(s) would you most likely need to use to find the length of a side? _____
12. Use your calculator to answer the following:
 2×41.587 _____ $2578/4$ _____ $369 + 578$ _____
➤ Now press 2^{nd} ENTER two times. What pops up on your screen? _____
➤ Arrow down and change the 4 to a 2. What answer do you get? _____

How will this feature be helpful? _____

- Go to **MODE** . On the second line, move your cursor to select option 2. Repeat the calculation 2×41.587 . Also calculate 41×20 . What does the second line affect? _____
- Change the second line back to float before continuing?

13. What happens when the **10^x** and **6** keys are pressed? _____

14. Press **MATH** , what do you think the first entry will do? _____

Now press **CLEAR** , then press **0** **.** **5** **6** **MATH** and select option 1 then **ENTER**

What answer do you get? _____

Is this what you expected? _____ Why? _____

15. The **STO→** button stores numbers to variables (among other things). To evaluate the expression $\frac{(2A+3B)}{(4-c)}$, when $A=9, B=2$ and $C=1$, first press **9** **STO→** **ALPHA** **MATH** **ENTER** to store the number 9

to A. Repeat this same process if $B=2$ and $C=1$, then evaluate the expression by typing in the expression $\frac{(2A+3B)}{(4-c)}$ (TI-84 Calculators have a the fraction button: Type **ALPHA** **y=** **1: n/d** which will bring up a

vertical fraction bar $\frac{\square}{\square}$ for your to type the expression) , then press **ENTER**.. What did you get? _____

Is it faster just to substitute the values into the expression and solve the old- fashioned way with paper and pencil? _____

When might this feature come in handy? _____

Evaluate $\frac{2A+3B}{4-c} =$ _____ Is there a difference between $\frac{2A+3B}{4-c}$ vs. $\frac{(2A+3B)}{(4-c)}$ on your calculator?

13. Press **2nd** **0** to access the calculator's catalogue. This list all of the commands your calculator recognizes. Scroll up, to access symbols.

What is the first symbol? _____. Press **MATH** **→** **1** **(-)** **4** . What did that do?

➤ Press **2nd** **0** to access the calculator's catalogue. What is the last symbol? _____

Did you scroll all the way to the bottom of the list? _____ It took a while didn't it? _____

14. Press **2nd** **0** to access the calculator's catalogue. Press the up arrow **↑** Describe what happened? _____ How does the catalogue appear to be organized? _____

➤ Press **2nd** **0** again ? What appear in the upper right hand corner? _____

This means the calculator is in alphabetical mode. Press **]**. What is the 5th entry in the L's? What do these letters stand for? _____

➤ Press **÷** . What is the 4th entry in the M's? _____

15. Press **4** **MATH**, choose option 5, then press **1** **6** and **ENTER**. What did this option do? _____

16. Which function allows you to send/receive data/programs? _____

17. Press **5** **÷** **÷** **9** **ENTER**. Press **2** to go to the error. The cursor should be blinking on the second /, press **DEL** **ENTER**. What answer did you get? _____ To convert this number to a fraction, press **MATH** **ENTER** and you get _____

18. Enter this problem into the calculator and press **ENTER**. $2.4 \times 3.7 =$ _____.

➤ Now press **MODE** **↓** Float **→** to 0 and press **ENTER**. Now press **2nd** **Quit** to return to the home screen and press **2nd** **ENTER** and the original problem should appear on the screen, now press **ENTER**. What appears on the screen? _____ Think about this number in relation to the answer you got before. What did the calculator do? _____

➤ Repeat this same process except select 3 under the Float option. Return to the home screen, recall the original problem and press **ENTER**. What number appears on the screen? _____ What did the calculator do this time? _____

19. Enter $(-2)^2$ into the calculator, what answer did you get? _____ Now enter -2^2 into the calculator, what answer did you get this time? _____ Why do you think you got two different answers? _____ Would $(-2)^3$ and -2^3 give you two different answers? Why or why not? _____

20. Press **Y=** type in type **2** **$\frac{Y}{T,q,n}$** **1**. Press **ZOOM** then select 6, press **MODE**, arrow to the bottom and arrow over to G-T and press **ENTER**. Now press **GRAPH**. What appears on the screen? _____

➤ Press **MODE** and scroll down to Full and press **ENTER** to restore to full screen.

➤ Press **GRAPH**, then press **TRACE**. What dot is blinking? _____

What is the x value on the bottom? _____

What is the y value on the bottom? _____

➤ Now press the left and right arrows. What happens to the blinking point?

➤ Press **TRACE** **6** **ENTER**. What do you get? _____

21. Press **2nd** **MODE** to go back to the home screen. Press **VARS** located next to clear button, press **→** to highlight Y-Vars, Press **1** for function, **1** for Y_{11} **6** (NOTE: TI-84 Calculators can call up Y_{11} also by selecting **ALPHA** **Trace** select Y_{11} enter) What do you get? _____

➤ Press **Y=** and press **↓** to go to Y_{12} and Press **VARS** located next to clear button, press **→** to highlight Y-Vars, Press **1** for function, **1** for Y_{11} **+** **3** **ENTER** **GRAPH**. What appears on the screen? _____

➤ How does $Y_{11} + 3$ affect the graph of Y_{11} ? _____

➤ When might Y-vars be helpful? _____

22. Press $\boxed{Y=}$ and press $\boxed{\downarrow}$ to go to Y_{13} and type $\boxed{5} \boxed{\sin} \boxed{\left(\frac{\pi}{4}T,q,n\right)} \boxed{+} \boxed{3}$, then move the blinking cursor over the $\boxed{=}$ in Y_{12} press $\boxed{\text{ENTER}}$ for each, What happens to the $\boxed{=}$ symbol?
 _____ now press $\boxed{\text{GRAPH}}$. What do you notice? _____.

➤ Press $\boxed{2^{\text{nd}}}$ $\boxed{\text{TRACE}}$ $\boxed{5}$ The blinking cursor is on what graph? _____. What is the equation in the upper left corner? _____, now Press $\boxed{\text{ENTER}}$, where is the blinking cursor now? _____ What equation is in the upper left corner? _____.
 Press $\boxed{\text{ENTER}}$ $\boxed{\text{ENTER}}$ (two times). What do you get? ? (Round your answer to 3 decimal places)

➤ Now Deselect the $\boxed{=}$ in Y_{11} press $\boxed{\text{ENTER}}$, $\boxed{\text{GRAPH}}$ so that you only see the curved graph

23. Press $\boxed{\text{ZOOM}}$ "7" $\boxed{\text{ZTrig}}$ What does that do to the scale on the graph? _____

➤ Can you see the high and low points on the graph? _____. Press $\boxed{\text{WINDOW}}$ and change the x- min to $-\rho$ and x-max to $\frac{\pi}{2}\rho$, Press $\boxed{\text{ZOOM}}$ "0" $\boxed{\text{ZoomFit}}$ $\boxed{\text{GRAPH}}$. What happens?

➤ Press $\boxed{\text{WINDOW}}$ and identify the x-scl? _____ What is the exact number for that decimal value? _____ What is the x-scl? _____

24. I hope you've had some fun, now press $\boxed{2^{\text{nd}}}$ $\boxed{\text{ON}}$ and we're done!